

James Dinh

Data Scientist & Engineer

 jamesdinh.me

 jamesdinh77@protonmail.com

 [jamesvdinh](https://github.com/jamesvdinh)

 [/in/jamesdinh-me/](https://in/jamesdinh-me/)

Experience

Data Analytics & UX Intern

Aug 2025 – present

Fung Institute for Engineering Leadership – UC Berkeley MEng Program

Berkeley, CA

- ❖ Cleaned, aggregated, and analyzed student pulse-survey data using Python and automated quality checks, producing reliable program metrics that informed Capstone Project resource allocation for 450+ MEng students
- ❖ Built an interactive keyword-frequency map component in React and D3.js that visualizes live data 225+ Capstone projects, enabling students to identify trending topics and inform their project rankings at a glance
- ❖ Developed a comprehensive MAS-E course-planning tool with automated prerequisite checks and forecasting logic, and integrated its data flows with Salesforce Data Cloud to unify advising and reporting across the program

Full Stack Software Engineer

Sep – Dec 2025

Prompt Opinion – AI Healthcare Copilot Platform

Remote (Contract)

- ❖ Engineered and integrated MCP tools and AI Agents in C# (.NET 9), implementing FHIR-compliant API workflows and reducing LLM-EHR data retrieval latency through optimized HTTP pipelines of patient data and Vue.js caching
- ❖ Developed and maintained full-stack features in close collaboration with Design and Product, using Vue.js, TypeScript, Vite, Node.js, and PostgreSQL; deployed to Azure via Docker and GitHub Actions CI/CD pipelines

AI & Data Science Intern

Jun – Aug 2025

Darena Health – AI-enabled, FHIR-first EHR

Remote

- ❖ Built an MCP server and data-integration pipeline connecting an LLM-powered clinical chatbot to an EHR, enabling real-time retrieval of patient FHIR data to power CVD risk scoring models and clinical trial matching
- ❖ Containerized and deployed the MCP server and data-integration pipeline using Docker and Kubernetes, providing consistent, isolated environments for bidirectional FHIR data exchange and improving stability for new features

Frontend Web Developer Intern

Aug 2023 – Aug 2024

Digital NEST – Web & Graphic Design Nonprofit

Gilroy, CA

- ❖ Coordinated with clients in education and recreation to produce, develop, and maintain websites using WordPress, JS, and PHP, implementing SEO best practices to boost page visits by 200% and sustain an average 36% bounce rate
- ❖ Integrated GCP APIs (Maps, Geocoding) in a park-finder feature, enabling users to find nearby parks by distance

Education

University of California, Berkeley

Expected Grad May 2026

B.A. Data Science

GPA: 3.85

- ❖ Coursework: ML, Data Engineering, Data Structures, Probability Theory, Numerical Analysis
- ❖ Concentration in Applied Math & Modeling; Clubs/Orgs: Cal VSA Lion Dance (Music Coordinator)

Gavilan College, Dual Enrollment with High School

Grad May 2023

A.S. Computer Programming | A.A. Natural Science

GPA: 3.95

Projects

MIMIC-IV-Ext Unsupervised Phenotyping of Diabetes Progression (Research Data Scientist)

Data Pipeline Utilizing K-Means & UMAP to Phenotype Diabetic Patients – [Research Paper](#)

- ❖ Designed and implemented a scalable ETL pipeline in Python to extract, clean, and transform heterogeneous clinical records across 46k+ diabetes patients from MIMIC-IV, engineering features for downstream unsupervised segmentation models (K-Means, UMAP) to identify clinically meaningful clusters
- ❖ Trained a PyTorch transformer to learn patient-level embeddings from heterogeneous, temporally irregular data

American Sign Language Vision-to-Text Translator (Full Stack Data Scientist)

CNN Model Trained on a Dataset of 87,000+ Gesture Images – [jamesvdinh.github.io/asl-translator](https://github.com/jamesvdinh/asl-translator)

- ❖ Implemented a feedforward neural network with 128-unit and 256-unit ReLU layers that classified inputs across 29 categories on a live camera feed input; Used binary cross-entropy loss & SGD to achieve an AUC of 0.97

Skills

- ❖ Python, SQL, PostgreSQL, TypeScript, JavaScript, React, Vue, Node.js, MongoDB, D3.js, Azure, C# .NET, Docker, Kubernetes, Git, GCP, ETL/ELT, REST API, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow
- ❖ Personal/Hobbies: Lion Dance, LEGOs, Boulderling, Videography, Guitar, Piano, Table Tennis, Chess